SOC 690S: Machine Learning in Causal Inference

Week 7: Instrumental Variable Estimation

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Department of Sociology, Fall 2025

Week	Date	Topic	Problem sets Assign	Problem sets Due
1	Aug 26	Introduction: Motivation and Linear Regression		
2	Sep 2	Foundation: ML-based Regression Regularization		
3	Sep 9	Foundation: Non-Linear Machine Learning	1	
4	Sep 16	Foundation: Double Machine Learning		
5	Sep 23	Foundation: Directed Acyclic Graph (DAG)	2	1
6	Sep 29	Core: Matching, Propensity, Weighting, and Beyond		
7	Oct 7	Core: Instrumental Variable Estimation	3	2
8	Oct 14	*Fall break*		
9	Oct 21	**In-class midterm**		
10	Oct 28	Core: Regression Discontinuity Design		
11	Nov 4	Core: Panel Data and Difference-in-Difference		3
12	Nov 11	Advanced: Heterogeneous Treatment Effect I	4	
13	Nov 18†	Advanced: Heterogeneous Treatment Effect II		
14	Nov 25	Advanced: Unstructured Data Feature Engineering		4
	Dec 13	**final project due**		

† Justin Lucas Sola, Assistant Professor of Sociology and Data Science at UNC, will join us to give a guest talk on heterogeneous treatment effects and their applications.

Instrumental Variable Estimation

Instrumental Variable Estimation Motivation

- Remember when we discussed DAG, we mentioned one scenario, where variables like Z are known as *instrumental* variables
- These variables can be thought as inducing natural experiments that can be leveraged for causal identification in the presence of unobserved confounding
- Instruments *should not* be used in an identification by adjustment strategy

$$
\begin{tikzcd}
Z \arrow[r] & S \arrow[r] & Y \\
& & A \arrow[ul] \arrow[u]
\end{tikzcd}
$$

Omitted Variable Bias

- Before going into the details of instrumental variable estimation, we start with the problem of *omitted variable bias*
- In the above case, for example, we do not observe latent variable U
- In estimating the effect of schooling S_i on wages Y_i, common latent variables include ability A_i
- The true data–generating process

$$Y_i = \alpha + \beta S_i + \gamma A_i + u_i$$

- where u_i is the remaining error with $E[u_i \mid A_i, S_i] = 0$
- We (mistakenly) estimate the *short regression* omitting A_i

$$Y_i = \alpha^A + \beta^A S_i + e_i$$

Omitted Variable Bias

- Suppose the population linear projection of A_i on S_i

$$A_i = \delta S_i + v_i, \qquad \text{with } E[S_i v_i] = 0,$$

where A_i is $k \times n$ matrix, and δ is a $k \times 1$ vector of slopes

- $\delta = \dfrac{Cov(A_i, S_i)}{V(S_i)}$ gives the slope(s) from regressing A_i on S_i

Omitted Variable Bias

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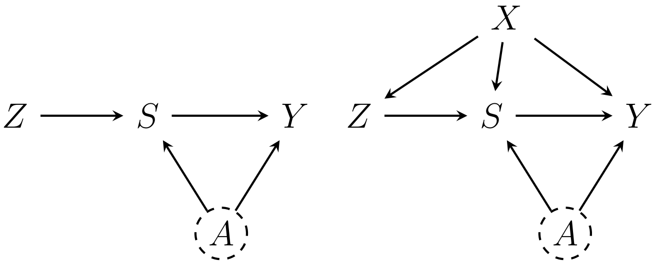
$$A_i = \delta S_i + v_i, \qquad \text{with } E[S_i v_i] = 0,$$
where A_i is $k \times n$ matrix, and δ is a $k \times 1$ vector of slopes
- $\delta = \dfrac{Cov(A_i, S_i)}{V(S_i)}$ gives the slope(s) from regressing A_i on S_i
- Substitute the projection into the true DGP
$$Y_i = \alpha + \beta S_i + \gamma'(\delta S_i + v_i) + u_i = \alpha + (\beta + \gamma'\delta)S_i + (\gamma' v_i + u_i)$$
- The *short regression* (of Y_i on S_i only) will pick up
$$\beta^A = \beta + \gamma'\delta$$
- Therefore, the *omitted variable bias (OVB)* is
$$\text{Bias}(\hat{\beta}^A) = \beta^A - \beta = \gamma'\delta = \gamma' \frac{Cov(A_i, S_i)}{V(S_i)}$$

Instrumental Variable Estimation Intuition

- OVB cannot be addressed by the methods we have covered so far, such as matching or propensity score weighting
- Leveraging *conditional ignorability* for causal identification assumes that we have complete set of X_i to control for
- Instrumental varaibles are introduced to "bypass" the unobservability of A_i



Basic Setup of Instrumental Variable Estimation

- In the most basic form, there are three variables, the instrument Z_i, the *endogenous* independent variable S_i, and the outcome variable Y_i

$$Y_i = \alpha + \beta S_i + u_i$$

- where S_i may be endogenous due to an unobserved confounder A_i

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- where S_i may be endogenous due to an unobserved confounder A_i
- The instrumental variable estimation relies on two key assumptions
- *Relevance: Z_i must shift S_i (non-zero first stage)*

$$\text{Cov}(Z_i, S_i) \neq 0$$

- *Exogeneity*

$$E[Z_i u_i] = 0$$

  Z_i affects Y_i only through S_i and is uncorrelated with latent A_i
- Observed characteristics X_i can be included to strengthen the two assumptions (*e.g.*, *conditional exogeneity:* $E[Z_i u_i \mid X_i] = 0$)

Basic Setup of Instrumental Variable Estimation

- Under these two assumptions, we can solve the *population* moment condition based on *exogeneity*

$$E\Big[Z_i(Y_i - \alpha - \beta S_i)\Big] = 0$$

- Using $\alpha = E[Y_i] - \beta E[S_i]$, the moment condition becomes

$$E[Z_iY_i] - E[Z_i]E[Y_i] - \beta\Big(E[Z_iS_i] - E[Z_i]E[S_i]\Big) = 0$$

- This is equivalent to

$$\beta = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, S_i)}$$

Intuition of the Ratio

- The numerator, $Cov(Z_i, Y_i)$, is the *reduced-form* association; how much Y_i changes with changes in the instrument Z_i
- The denominator, $Cov(Z_i, S_i)$ is the *first-stage* association; how much the endogenous regressor S_i moves with Z_i
- The IV estimand

$$\beta = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, S_i)} = \frac{Cov(Z_i, Y_i)/V(Z_i)}{Cov(Z_i, S_i)/V(Z_i)}$$

$$\begin{tikzcd}
Z \arrow[r, "\gamma"] & S \arrow[r, "\beta"] & Y \\
& A \arrow[u] \arrow[ur] &
\end{tikzcd}$$

The Colonial Origins of Comparative Development

- The canonical paper *The Colonial Origins of Comparative Development* by Acemoglu, Johnson, and Robinson (2001) estimates the effect of institutions on economic performance
- The form of institution measured by *e.g.*, protection from expropriation risk based on the International Country Risk Guide (ICRG) can be *endogeneous* to economic performance
- AJR used historical settler mortality to instrument institutional quality

The Colonial Origins of Comparative Development

- European colonizers adopted different institutions depending on settlement feasibility
- In *low mortality* areas, settlers stayed and created inclusive institutions such as property rights and rule of law
- In *high mortality* areas, few settlers stayed and colonization happened mainly through establishing extractive institutions for resource extraction
- AJR constructed settler mortality from historical materials and presented the *reduced-form* association

The Colonial Origins of Comparative Development

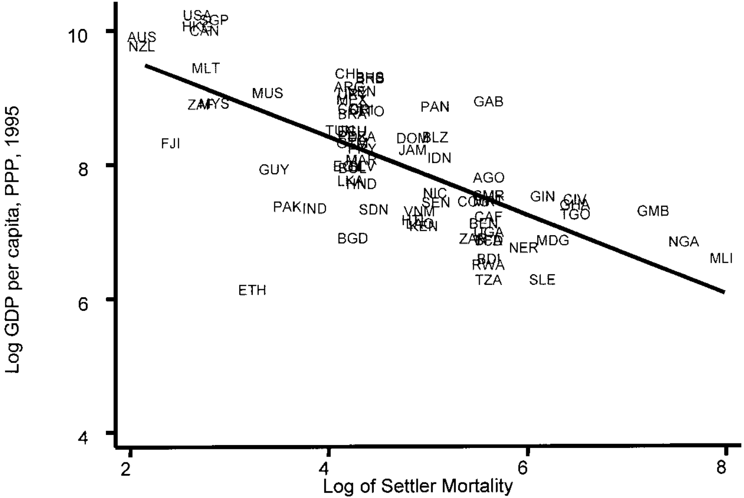

Figure: Reduced-Form Relation Between Income and Settler Mortality

The Colonial Origins of Comparative Development

- AJR noted that their IV strategy will be invalidated if other factors that affect settler mortality rate are also persistent and related to the development of institutions within a country and to the country's GDP
- A leading candidate is *geography*
- AJR addressed it by assuming that the confounding effect of *geography* is adequately captured by a linear term in distance from the equator and a set of continent dummy variables
- We will showcase how machine learning can relatex the linearity assumption and include high-dimensional controls in this classic low-dimensional cross-country analysis

Quarter-of-Birth as an Instrument

- Angrist and Krueger (1991) instrumented years of schooling S_i using the combination of school start-age policies and compulsory schooling laws, under which children are compelled to attend school for different lengths of time, depending on their birthdays
- Most US states require children to start school in the calendar year they turn 6
- Children born late in the year (Q4) start school *younger* than those born early (Q1)
- Compulsory schooling laws require students to stay in school until age 16, such that students born in different quarters reach the legal dropout age after staying in school for different lengths of time

Quarter-of-Birth as an Instrument

- The two assumptions are likely satisfied
- Quarter of birth (Z_i) affects years of schooling (S_i) through institutional rules
- Quarter of birth (Z_i) is likely independent of unobserved innate ability (A_i)
- The policy-induced timing of school entry creates random-like variation in education length across individuals
- In Angrist and Krueger (1991), they controlled for age, year of birth, and state of birth as additional controls X_i

Quarter-of-Birth as an Instrument: First-Stage

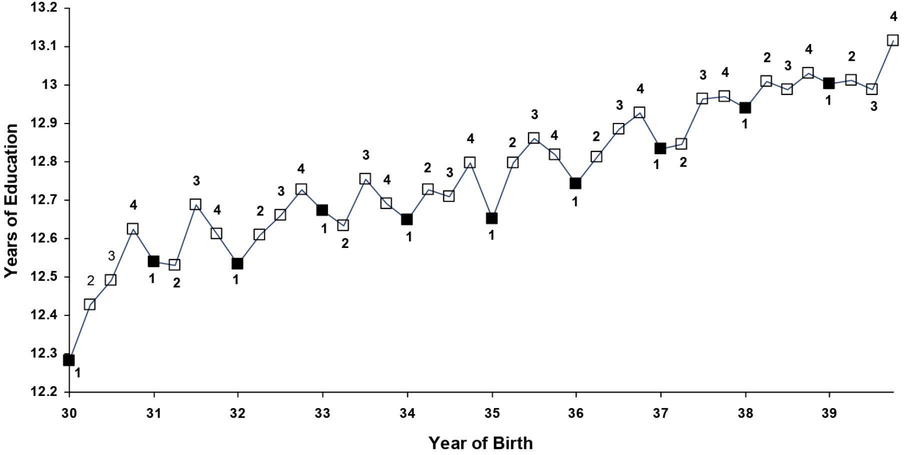

Figure: Average Education by Quarter of Birth (first stage)

Quarter-of-Birth as an Instrument: Reduced Form

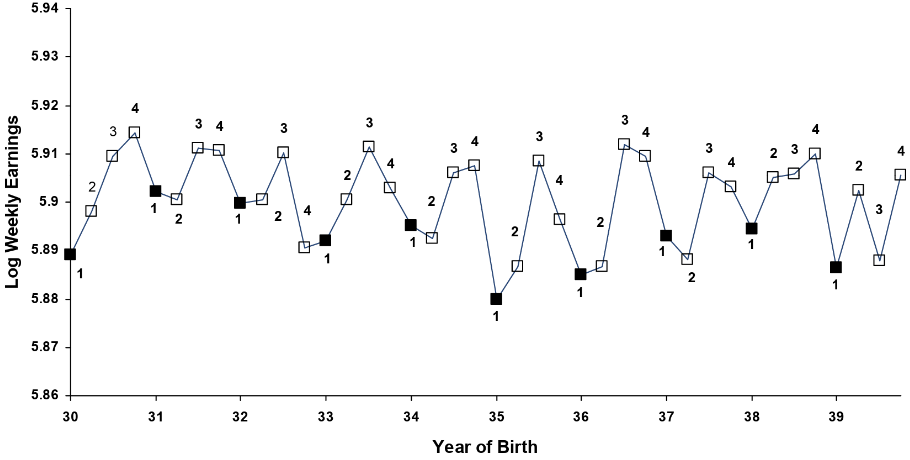

Figure: Average Weekly Wage by Quarter of Birth (reduced form)

Two-Stage Least Squares

- IV estimation can be performed through Two-Stage Least Squares (2SLS)
- Angrist and Krueger (1991) setup can be expressed in two stages

$$S_i = \pi_{10}' X_i + \pi_{11} Z_i + \epsilon_{1i}$$
$$Y_i = \pi_{20}' X_i + \pi_{21} Z_i + \epsilon_{2i}$$

- To estimate the causal model with covariates

$$Y_i = \alpha' X_i + \beta S_i + u_i$$

- β can be recovered by

$$\beta = \frac{\pi_{21}}{\pi_{11}} = \frac{Cov(Y_i, \tilde{Z}_i)}{Cov(S_i, \tilde{Z}_i)} = \frac{Cov(\tilde{Y}_i, \tilde{Z}_i)}{Cov(\tilde{S}_i, \tilde{Z}_i)}$$

Two-Stage Least Squares

- The reduced-form equation can be derived by substituting the first-stage equation into the causal model

$$\begin{aligned} Y_i &= \alpha' X_i + \beta(\pi_{10}' X_i + \pi_{11} Z_i + \epsilon_{1i}) + u_i \\ &= (\alpha + \beta\pi_{10})' X_i + \beta\pi_{11} Z_i + \beta\epsilon_{1i} + u_i \end{aligned}$$

- where $\pi_{20} \equiv \alpha + \beta\pi_{10}$, $\pi_{21} \equiv \beta\pi_{11}$
- It again shows why $\beta = \pi_{21}/\pi_{11}$
- Slightly rearranging the equation

$$Y_i = \alpha' X_i + \beta(\pi_{10}' X_i + \pi_{11} Z_i) + \epsilon_2$$

- where $(\pi_{10}' X_i + \pi_{11} Z_i)$ is the population fitted value from the first-stage regression of S_i on X_i and Z_i
- As Z_i and X_i are uncorrelated with the reduced form error ϵ_{2i}, the coefficient on $(\pi_{10}' X_i + \pi_{11} Z_i)$ in the population regression of Y_i on X_i and $(\pi_{10}' X_i + \pi_{11} Z_i)$ equals β

Two-Stage Least Squares

- Given a random sample, the first-stage fitted values are estimated by

$$\hat{S}_i = \hat{\pi}_{10}' X_i + \hat{\pi}_{11} Z_i$$

- where $\hat{\pi}_{10}$ and $\hat{\pi}_{11}$ are OLS estimates
- $\hat{\beta}$ is estimated by

$$Y_i = \alpha' X_i + \beta \hat{S}_i + [u_i + \beta(S_i - \hat{S}_i)]$$

Standard Error and Weak Instrument

IV Estimation in Matrix Form

- The 2SLS is correct in point estimation, but *incorrect* in standard error estimation
- The standard heteroskedascitiy *sandwich* estimator is however straightforward
- Suppose a simplified model with one endogenous regressor and k-dimensional instrument $Z \in \mathbb{R}^{n \times k}$, with the main causal model as

$$Y = D\beta_0 + u$$

- From the moment condition $E[Z'u] = 0$

$$E[Z'(Y - D\beta)] = 0$$

IV Estimation in Matrix Form

- The sample analog

$$
\begin{aligned}
Z'(Y - D\hat{\beta}) &= 0 \\
(Z'D)\hat{\beta} &= Z'Y \\
D'Z(Z'Z)^{-1}(Z'D)\hat{\beta} &= D'Z(Z'Z)^{-1}Z'Y \\
\hat{\beta} &= (D'Z(Z'Z)^{-1}Z'D)^{-1}D'Z(Z'Z)^{-1}Z'Y
\end{aligned}
$$

- More compactly, define the idempotent and symmetric projection matrix $P_Z = Z(Z'Z)^{-1}Z'$

$$\hat{\beta} = (D'P_ZD)^{-1}D'P_ZY$$

- Note here P_ZD is first-stage fitted value; from 2SLS, $\hat{\beta}$ can be intuitively derived by

$$\hat{\beta} = ((P_ZD)'(P_ZD))^{-1}(P_ZD)'Y = (D'P_ZD)^{-1}D'P_ZY$$

Sandwich Estimator of variance in IV

- To derive the variance of $\hat{\beta}$, substitute $Y = D\beta_0 + u$ into the IV formula

$$\hat{\beta} - \beta_0 = (D'P_Z D)^{-1} D'P_Z(D\beta_0 + u) - \beta_0 = (D'P_Z D)^{-1} D'P_Z u$$

- Multiply the equation by $\sqrt{n}$ for asymptotics; under regularity and *strong* instrument

$$\frac{1}{n} D'P_Z D \xrightarrow{p} A > 0, \qquad \frac{1}{\sqrt{n}} D'P_Z u \xrightarrow{d} N(0, S)$$

- By *Slutsky Theorem*

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, A^{-1} S A^{-1})$$

- Such that the robust sandwich estimator of the asymptotic variance

$$V(\hat{\beta}) = (D'P_Z D)^{-1}\ D'P_Z V(u_i) P_Z D\ (D'P_Z D)^{-1}$$

In practice plug-in $\hat{V}(u_i) = \text{diag}(\hat{u}_i^2)$ to get the robust SE

Weak Instruments: Variance Inflation and Normality Failure

- Variance estimation depends on the magnitude of the denominator matrix $(D'P_Z D)^{-1}$
- If instruments explain little of D, that matrix is small and its inverse is large
- While details are omitted, it is established that under a weak instrument regime, the variance of $\hat{\beta}$ is *not* asymptotically normal; it has a non-normal, heavy-tailed limit (no Gaussian limit with finite variance), and standard t-test fails

Diagnosing Weakness: the First-Stage F-Statistic

- From the first-stage regression, compute the F-statistics testing the null hypothesis that there is no association between the instrument and the *endogenous* variable
- Rule of thumb for strong instrument (Stock-Yogo, 2005):

$$F \geq 10$$

Anderson-Rubin (AR) Approach to Weak Instrument

- *Anderson–Rubin (AR)* directly tests the moment condition $E[(Y - \theta D)Z] = 0$ and yields valid CIs even under weak instruments. (Anderson and Rubin, 1949)
- The idea is instead of using the inverse matrix $(D'P_Z D)^{-1}$, one may search $\hat{\theta}$ and include the range of possible $\hat{\theta}$ in the confidence interval until $E[(Y - \theta D)Z] \neq 0$ cannot be rejected at a prespecified confidence level

Anderson-Rubin (AR) Approach to Weak Instrument

- AR defines the statistic

$$C(\theta) = \frac{|\mathbb{E}_n[(\tilde{Y} - \theta\tilde{D})\tilde{Z}]|^2}{\mathbb{V}_n[(\tilde{Y} - \theta\tilde{D})\tilde{Z}]/n}$$

- If $\theta_0 = \theta$, then $C(\theta) \xrightarrow{d} \mathcal{N}(0,1)^2 = \chi^2(1)$
- We can reject the hypothesis $\theta_0 = \theta$ at level a (for example $a = 0.05$ for a 5% level test) if $C(\theta) > c(1-a)$ where $c(1-a)$ is the $(1-a)$- quantile of a $\chi^2(1)$ variable
- The probability of falsely rejecting the true hypothesis is approximately $a \times 100\%$
- To construct a $(1-a) \times 100\%$ confidence region for θ, we can then simply invert the test by collecting all parameter values that are not rejected at the a level

$$CR(\theta) = \{\theta \in \Theta : C(\theta) \leq c(1-a)\}$$

The Wald Estimator

The Wald Estimator with a Binary Instrument

- We consider a particular scenario where we have a binary instrument Z_i, binary treatment D_i, outcome Y_i
- We assume random assignment of Z_i (*independence*):

$$Z_i \perp \{Y_i(1), Y_i(0), D_i(1), D_i(0)\}$$

- *Exogeneity*: Z_i affects Y_i only through D_i: $Y_i(z, d) = Y_i(d)$
- *Relevance*: $E[D_i(1) - D_i(0)] \neq 0$.
- *Monotonicity*: $D_i(1) \geq D_i(0)$ for all i (*i.e.*, no defiers)

The Wald Estimator with a Binary Treatment

- The *population* Wald estimator is

$$\beta_{Wald} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

- We want to show that Wald estimator, and more broadly, the IV strategy, speaks to the treatment effect only for compliers

Wald Estimates of the Effect of Military Service on Earnings

Table 4.1.3: Wald estimates of the effects of military service on the earnings of white men born in 1950

	Earnings		Veteran Status		Wald Estimate of Veteran Effect
Earnings year	Mean	Eligibility Effect	Mean	Eligibility Effect	
	(1)	(2)	(3)	(4)	(5)
1981	16,461	-435.8	0.267	0.159	-2,741
		(210.5)		(0.040)	(1,324)
1971	3,338	-325.9			-2050
		(46.6)			(293)
1969	2,299	-2.0			
		(34.5)			

The Wald Estimator with a Binary Treatment

- For any unit i, if $Z_i = 1$ we observe $Y_i = Y_i(D_i(1))$. Therefore

$$E[Y_i \mid Z_i = 1] = E\Big[Y_i(D_i(1)) \mid Z_i = 1\Big]$$

- By independence ($Z \perp$ potential outcomes and potential treatments), conditioning on $Z_i = 1$ does not change the distribution of the vector $\{Y_i(1), Y_i(0), D_i(1), D_i(0)\}$. Hence

$$E[Y_i(D_i(1)) \mid Z_i = 1] = E[Y_i(D_i(1))]$$
$$E[Y_i \mid Z_i = 1] = E\Big[Y_i(D_i(1))\Big]$$

- Analogously, $E[Y_i \mid Z_i = 0] = E[Y_i(D_i(0))]$

The Wald Estimator with a Binary Treatment

- Start with the difference

$$E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] \;=\; E\Big[Y_i(D_i(1)) - Y_i(D_i(0))\Big]$$

- Expand each $Y_i(D_i(z))$ using the definition of potential outcomes

$$\begin{aligned} Y_i(D_i(1)) &= Y_i(1)\cdot D_i(1) \;+\; Y_i(0)\cdot(1 - D_i(1)) \\ Y_i(D_i(0)) &= Y_i(1)\cdot D_i(0) \;+\; Y_i(0)\cdot(1 - D_i(0)) \end{aligned}$$

- Substitute into the difference:

$$\begin{aligned} Y_i(D_i(1)) - Y_i(D_i(0)) = &\Big[Y_i(1)D_i(1) + Y_i(0)(1 - D_i(1))\Big] \\ &- \Big[Y_i(1)D_i(0) + Y_i(0)(1 - D_i(0))\Big] \end{aligned}$$

The Wald Estimator with a Binary Treatment

- Continue simplifying the bracketed expression

$$
\begin{aligned}
& Y_i(1)D_i(1) + Y_i(0)(1 - D_i(1)) - Y_i(1)D_i(0) - Y_i(0)(1 - D_i(0)) \\
&= Y_i(1)D_i(1) - Y_i(1)D_i(0) \ + \ Y_i(0)(1 - D_i(1)) - Y_i(0)(1 - D_i(0)) \\
&= Y_i(1)\left[D_i(1) - D_i(0)\right] \ + \ Y_i(0)\left[(1 - D_i(1)) - (1 - D_i(0))\right] \\
&= Y_i(1)\left[D_i(1) - D_i(0)\right] \ + \ Y_i(0)\left[-D_i(1) + D_i(0)\right] \\
&= \Big[Y_i(1) - Y_i(0)\Big]\left[D_i(1) - D_i(0)\right].
\end{aligned}
$$

- Therefore, taking expectations,

$$
E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] = E\Big[(Y_i(1) - Y_i(0))\,(D_i(1) - D_i(0))\Big]
$$

The Wald Estimator with a Binary Treatment

- Denominator is

$$E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0] = E[D_i(1)] - E[D_i(0)] = E[D_i(1) - D_i(0)].$$

- Therefore, the Wald estimator

$$\beta_{\text{Wald}} = \frac{E[(Y_i(1) - Y_i(0))(D_i(1) - D_i(0))]}{E[D_i(1) - D_i(0)]}$$

The Wald Estimator with a Binary Treatment

- There are four types of units classified by the pair $(D_i(1), D_i(0))$

Complier:	$(1, 0)$
Always-taker:	$(1, 1)$
Never-taker:	$(0, 0)$
Defier:	$(0, 1)$

- Define indicator functions

$$C_i = \mathbb{1}\{D_i(1) = 1, D_i(0) = 0\} \quad A_i = \mathbb{1}\{D_i(1) = 1, D_i(0) = 1\}$$

$$N_i = \mathbb{1}\{D_i(1) = 0, D_i(0) = 0\} \quad F_i = \mathbb{1}\{D_i(1) = 0, D_i(0) = 1\}$$

- These indicators partition the population: $C_i + A_i + N_i + F_i = 1$

The Wald Estimator with a Binary Treatment

- Evaluate the difference $D(1) - D(0)$ by type

$$\begin{array}{ll} \text{Complier:} & D_i(1) - D_i(0) = 1 \\ \text{Always-taker:} & D_i(1) - D_i(0) = 0 \\ \text{Never-taker:} & D_i(1) - D_i(0) = 0 \\ \text{Defier:} & D_i(1) - D_i(0) = -1 \end{array}$$

- Pointwise, we write

$$D_i(1) - D_i(0) = C - F$$

The Wald Estimator with a Binary Treatment

- The Wald estimator is equivalent to

$$\beta_{wald} = \frac{E[(Y(1)-Y(0))C] \ - \ E[(Y(1)-Y(0))F]}{E[C] \ - \ E[F]}$$

- This is a weighted average of individual effects, with the defiers entering with a negative weight
- If we assume *monotonicity* ($E[F] = 0$)

$$\beta_{wald} = \frac{E[(Y_i(1)-Y_i(0))C]}{E[C]} = E\Big[\,Y_i(1) - Y_i(0) \mid \text{complier}\,\Big]$$

- Which is exactly the *Local Average Treatment Effect (LATE)*

$$\beta_{wald} = E[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)]$$